

2015 HIGHER SCHOOL CERTIFICATE EXAMINATION

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–18

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Which conic has eccentricity $\frac{\sqrt{13}}{3}$?

(A) $\frac{x^2}{3} + \frac{y^2}{2} = 1$

(B) $\frac{x^2}{3^2} + \frac{y^2}{2^2} = 1$

(C) $\frac{x^2}{3} - \frac{y^2}{2} = 1$

(D) $\frac{x^2}{3^2} - \frac{y^2}{2^2} = 1$

2 What value of z satisfies $z^2 = 7 - 24i$?

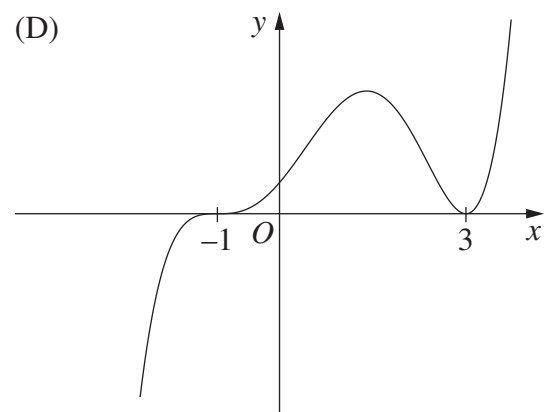
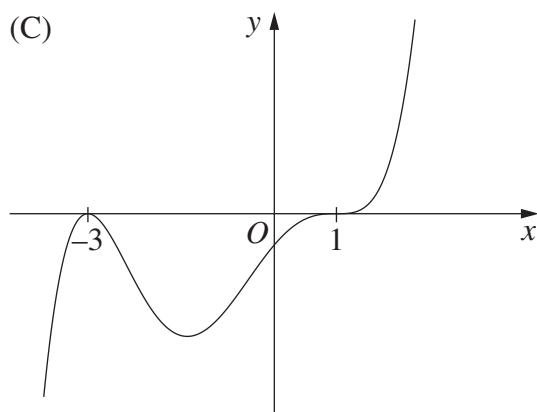
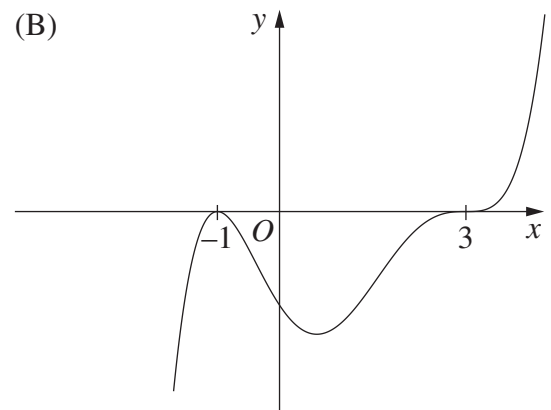
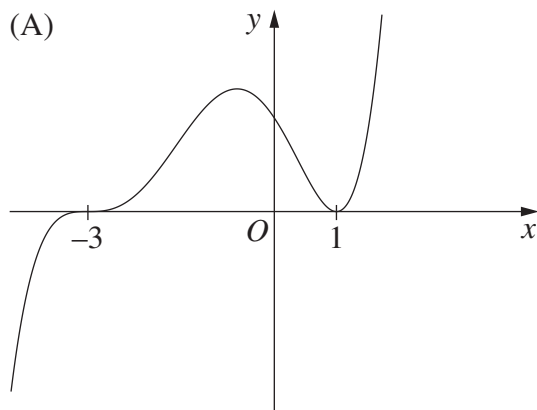
(A) $4 - 3i$

(B) $-4 - 3i$

(C) $3 - 4i$

(D) $-3 - 4i$

3 Which graph best represents the curve $y = (x - 1)^2(x + 3)^5$?



4 The polynomial $x^3 + x^2 - 5x + 3$ has a double root at $x = \alpha$.

What is the value of α ?

(A) $-\frac{5}{3}$

(B) -1

(C) 1

(D) $\frac{5}{3}$

5 Given that $z = 1 - i$, which expression is equal to z^3 ?

(A) $\sqrt{2} \left(\cos\left(\frac{-3\pi}{4}\right) + i \sin\left(\frac{-3\pi}{4}\right) \right)$

(B) $2\sqrt{2} \left(\cos\left(\frac{-3\pi}{4}\right) + i \sin\left(\frac{-3\pi}{4}\right) \right)$

(C) $\sqrt{2} \left(\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right)$

(D) $2\sqrt{2} \left(\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right)$

6 Which expression is equal to $\int x^2 \sin x \, dx$?

(A) $-x^2 \cos x - \int 2x \cos x \, dx$

(B) $-2x \cos x + \int x^2 \cos x \, dx$

(C) $-x^2 \cos x + \int 2x \cos x \, dx$

(D) $-2x \cos x - \int x^2 \cos x \, dx$

7 The numbers $1, 2, \dots, n$, for $n \geq 4$, are randomly arranged in a row.

What is the probability that the number 1 is somewhere to the left of the number 2?

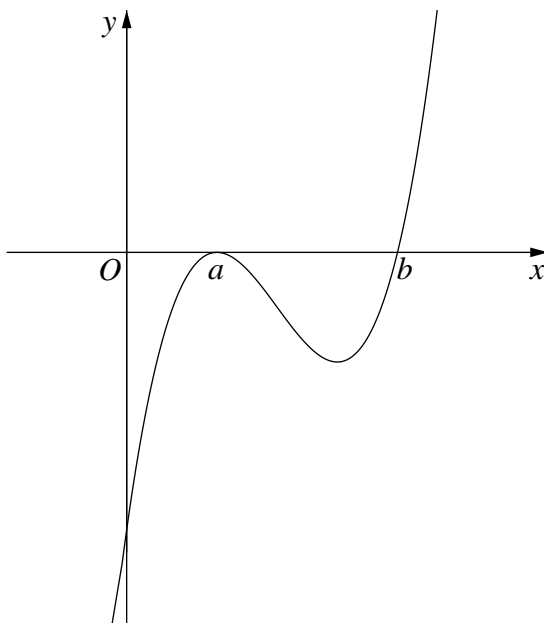
(A) $\frac{1}{2}$

(B) $\frac{1}{n}$

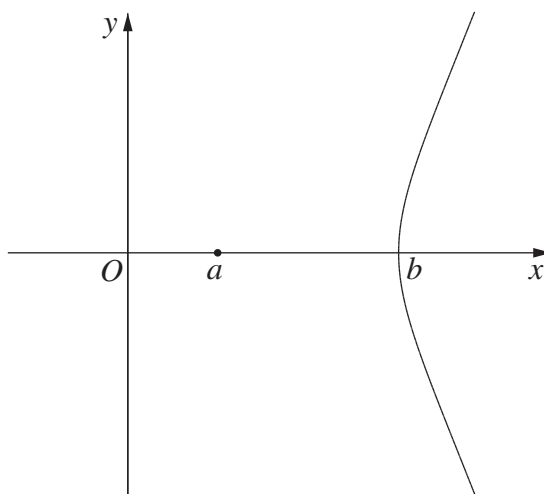
(C) $\frac{1}{2(n-2)!}$

(D) $\frac{1}{2(n-1)!}$

- 8 The graph of the function $y = f(x)$ is shown.



A second graph is obtained from the function $y = f(x)$.



Which equation best represents the second graph?

- (A) $y^2 = |f(x)|$
- (B) $y^2 = f(x)$
- (C) $y = \sqrt{f(x)}$
- (D) $y = f(\sqrt{x})$

- 9 The complex number z satisfies $|z - 1| = 1$.

What is the greatest distance that z can be from the point i on the Argand diagram?

- (A) 1
- (B) $\sqrt{5}$
- (C) $2\sqrt{2}$
- (D) $\sqrt{2} + 1$

- 10 Consider the expansion of

$$(1 + x + x^2 + \cdots + x^n)(1 + 2x + 3x^2 + \cdots + (n+1)x^n).$$

What is the coefficient of x^n when $n = 100$?

- (A) 4950
- (B) 5050
- (C) 5151
- (D) 5253

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Express $\frac{4+3i}{2-i}$ in the form $x+iy$, where x and y are real. **2**
- (b) Consider the complex numbers $z = -\sqrt{3} + i$ and $w = 3\left(\cos\frac{\pi}{7} + i\sin\frac{\pi}{7}\right)$.
- (i) Evaluate $|z|$. **1**
- (ii) Evaluate $\arg(z)$. **1**
- (iii) Find the argument of $\frac{z}{w}$. **1**
- (c) Find A , B and C such that $\frac{1}{x(x^2+2)} = \frac{A}{x} + \frac{Bx+C}{x^2+2}$. **2**
- (d) Sketch $\frac{x^2}{25} + \frac{y^2}{16} = 1$ indicating the coordinates of the foci. **2**
- (e) Find the value of $\frac{dy}{dx}$ at the point $(2, -1)$ on the curve $x + x^2y^3 = -2$. **3**
- (f) (i) Show that $\cot\theta + \operatorname{cosec}\theta = \cot\left(\frac{\theta}{2}\right)$. **2**
- (ii) Hence, or otherwise, find $\int (\cot\theta + \operatorname{cosec}\theta) d\theta$. **1**

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) The complex number z is such that $|z| = 2$ and $\arg(z) = \frac{\pi}{4}$.

Plot each of the following complex numbers on the same half-page Argand diagram.

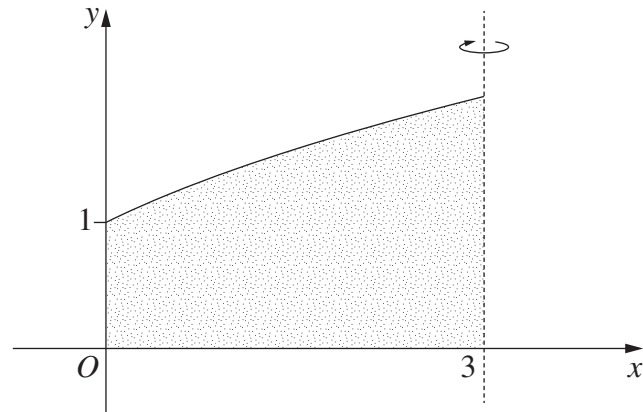
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|-------|---------------------|----------|
| (i) | z | 1 |
| (ii) | $u = z^2$ | 1 |
| (iii) | $v = z^2 - \bar{z}$ | 1 |
- (b) The polynomial $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$ has roots $a + ib$ and $a + 2ib$ where a and b are real and $b \neq 0$.
- | | | |
|------|--|----------|
| (i) | By evaluating a and b , find all the roots of $P(x)$. | 3 |
| (ii) | Hence, or otherwise, find one quadratic polynomial with real coefficients that is a factor of $P(x)$. | 1 |
- (c) (i) By writing $\frac{(x-2)(x-5)}{x-1}$ in the form $mx + b + \frac{a}{x-1}$, find the equation of the oblique asymptote of $y = \frac{(x-2)(x-5)}{x-1}$.
- | | | |
|------|---|----------|
| (ii) | Hence sketch the graph $y = \frac{(x-2)(x-5)}{x-1}$, clearly indicating all intercepts and asymptotes. | 2 |
|------|---|----------|

Question 12 continues on page 9

Question 12 (continued)

- (d) The diagram shows the graph $y = \sqrt{x+1}$ for $0 \leq x \leq 3$. The shaded region is rotated about the line $x = 3$ to form a solid.

4



Use the method of cylindrical shells to find the volume of the solid.

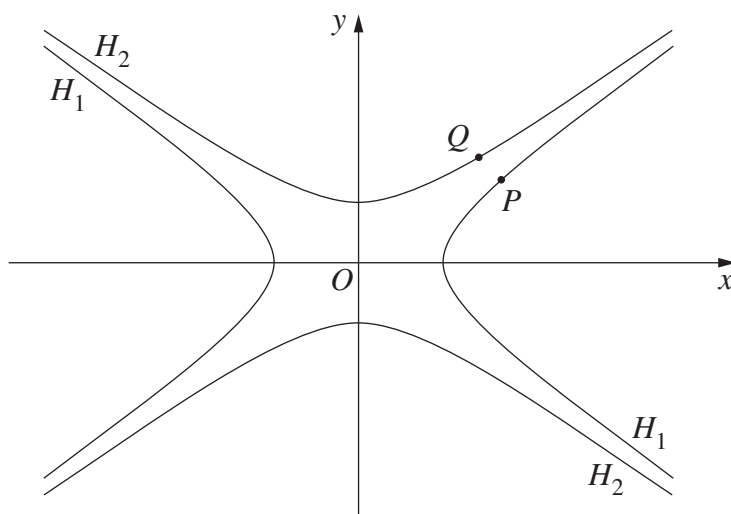
End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The hyperbolas $H_1 : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $H_2 : \frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ are shown in the diagram.

Let $P(a \sec \theta, b \tan \theta)$ lie on H_1 as shown on the diagram.

Let Q be the point $(a \tan \theta, b \sec \theta)$.

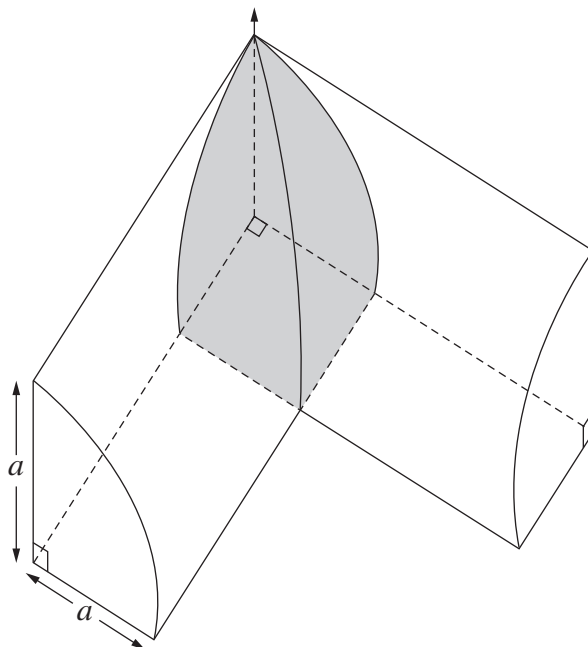


- | | |
|---|----------|
| (i) Verify that the coordinates of $Q(a \tan \theta, b \sec \theta)$ satisfy the equation for H_2 . | 1 |
| (ii) Show that the equation of the line PQ is $bx + ay = ab(\tan \theta + \sec \theta)$. | 2 |
| (iii) Prove that the area of $\triangle OPQ$ is independent of θ . | 3 |

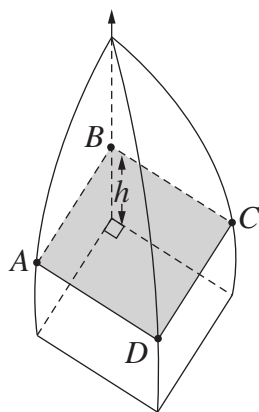
Question 13 continues on page 11

Question 13 (continued)

- (b) Two quarter cylinders, each of radius a , intersect at right angles to form the shaded solid.



A horizontal slice $ABCD$ of the solid is taken at height h from the base. You may assume that $ABCD$ is a square, and is parallel to the base.



- | | |
|---|----------|
| (i) Show that $AB = \sqrt{a^2 - h^2}$. | 1 |
| (ii) Find the volume of the solid. | 2 |

Question 13 continues on page 12

Question 13 (continued)

- (c) A small spherical balloon is released and rises into the air. At time t seconds, it has radius r cm, surface area $S = 4\pi r^2$ and volume $V = \frac{4}{3}\pi r^3$.

As the balloon rises it expands, causing its surface area to increase at a rate of $\left(\frac{4\pi}{3}\right)^{\frac{1}{3}} \text{ cm}^2 \text{ s}^{-1}$. As the balloon expands it maintains a spherical shape.

- (i) By considering the surface area, show that $\frac{dr}{dt} = \frac{1}{8\pi r} \left(\frac{4}{3}\pi\right)^{\frac{1}{3}}$. **2**
- (ii) Show that $\frac{dV}{dt} = \frac{1}{2}V^{\frac{1}{3}}$. **2**
- (iii) When the balloon is released its volume is 8000 cm^3 . When the volume of the balloon reaches $64\,000 \text{ cm}^3$ it will burst. **2**

How long after it is released will the balloon burst?

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Differentiate $\sin^{n-1}\theta \cos \theta$, expressing the result in terms of $\sin \theta$ only. **2**

(ii) Hence, or otherwise, deduce that $\int_0^{\frac{\pi}{2}} \sin^n \theta d\theta = \frac{(n-1)}{n} \int_0^{\frac{\pi}{2}} \sin^{n-2} \theta d\theta$, **2**
for $n > 1$.

(iii) Find $\int_0^{\frac{\pi}{2}} \sin^4 \theta d\theta$. **1**

(b) The cubic equation $x^3 - px + q = 0$ has roots α , β and γ .

It is given that $\alpha^2 + \beta^2 + \gamma^2 = 16$ and $\alpha^3 + \beta^3 + \gamma^3 = -9$.

(i) Show that $p = 8$. **1**

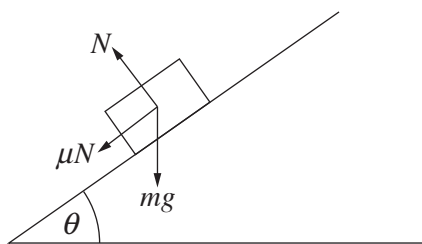
(ii) Find the value of q . **2**

(iii) Find the value of $\alpha^4 + \beta^4 + \gamma^4$. **2**

Question 14 continues on page 14

Question 14 (continued)

- (c) A car of mass m is driven at speed v around a circular track of radius r . The track is banked at a constant angle θ to the horizontal, where $0 < \theta < \frac{\pi}{2}$. At the speed v there is a tendency for the car to slide up the track. This is opposed by a frictional force μN , where N is the normal reaction between the car and the track, and $\mu > 0$. The acceleration due to gravity is g .



- (i) Show that $v^2 = rg \left(\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right)$. 3
- (ii) At the particular speed V , where $V^2 = rg$, there is still a tendency for the car to slide up the track. 2

Using the result from part (i), or otherwise, show that $\mu < 1$.

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) A particle A of unit mass travels horizontally through a viscous medium. When $t = 0$, the particle is at point O with initial speed u . The resistance on particle A due to the medium is kv^2 , where v is the velocity of the particle at time t and k is a positive constant.

When $t = 0$, a second particle B of equal mass is projected vertically upwards from O with the same initial speed u through the same medium. It experiences both a gravitational force and a resistance due to the medium. The resistance on particle B is kw^2 , where w is the velocity of the particle B at time t . The acceleration due to gravity is g .

- (i) Show that the velocity v of particle A is given by $\frac{1}{v} = kt + \frac{1}{u}$. **2**

- (ii) By considering the velocity w of particle B , show that **3**

$$t = \frac{1}{\sqrt{gk}} \left(\tan^{-1} \left(u \sqrt{\frac{k}{g}} \right) - \tan^{-1} \left(w \sqrt{\frac{k}{g}} \right) \right).$$

- (iii) Show that the velocity V of particle A when particle B is at rest is given by **1**

$$\frac{1}{V} = \frac{1}{u} + \sqrt{\frac{k}{g}} \tan^{-1} \left(u \sqrt{\frac{k}{g}} \right).$$

- (iv) Hence, if u is very large, explain why $V \approx \frac{2}{\pi} \sqrt{\frac{g}{k}}$. **1**

Question 15 continues on page 16

Question 15 (continued)

(b) Suppose that $x \geq 0$ and n is a positive integer.

(i) Show that $1 - x \leq \frac{1}{1+x} \leq 1$. **2**

(ii) Hence, or otherwise, show that $1 - \frac{1}{2n} \leq n \ln \left(1 + \frac{1}{n} \right) \leq 1$. **2**

(iii) Hence, explain why $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$. **1**

(c) For positive real numbers x and y , $\sqrt{xy} \leq \frac{x+y}{2}$. (Do NOT prove this.)

(i) Prove $\sqrt{xy} \leq \sqrt{\frac{x^2 + y^2}{2}}$, for positive real numbers x and y . **1**

(ii) Prove $\sqrt[4]{abcd} \leq \sqrt{\frac{a^2 + b^2 + c^2 + d^2}{4}}$, for positive real numbers a, b, c and d . **2**

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) A table has 3 rows and 5 columns, creating 15 cells as shown. 2

Counters are to be placed randomly on the table so that there is one counter in each cell. There are 5 identical black counters and 10 identical white counters.

Show that the probability that there is exactly one black counter in each column is $\frac{81}{1001}$.

- (ii) The table is extended to have n rows and q columns. There are nq counters, where q are identical black counters and the remainder are identical white counters. The counters are placed randomly on the table with one counter in each cell. 2

Let P_n be the probability that each column contains exactly one black counter.

Show that $P_n = \frac{n^q}{\binom{nq}{q}}$.

- (iii) Find $\lim_{n \rightarrow \infty} P_n$. 2

Question 16 continues on page 18

Question 16 (continued)

(b) Let n be a positive integer.

(i) By considering $(\cos \alpha + i \sin \alpha)^{2n}$, show that 2

$$\begin{aligned} \cos(2n\alpha) &= \cos^{2n}\alpha - \binom{2n}{2}\cos^{2n-2}\alpha \sin^2\alpha + \binom{2n}{4}\cos^{2n-4}\alpha \sin^4\alpha - \cdots \\ &\quad + \cdots + (-1)^{n-1}\binom{2n}{2n-2}\cos^2\alpha \sin^{2n-2}\alpha + (-1)^n \sin^{2n}\alpha. \end{aligned}$$

Let $T_{2n}(x) = \cos(2n \cos^{-1} x)$, for $-1 \leq x \leq 1$.

(ii) Show that 2

$$T_{2n}(x) = x^{2n} - \binom{2n}{2}x^{2n-2}(1-x^2) + \binom{2n}{4}x^{2n-4}(1-x^2)^2 + \cdots + (-1)^n(1-x^2)^n.$$

(iii) By considering the roots of $T_{2n}(x)$, find the value of 3

$$\cos\left(\frac{\pi}{4n}\right)\cos\left(\frac{3\pi}{4n}\right)\cdots\cos\left(\frac{(4n-1)\pi}{4n}\right).$$

(iv) Prove that 2

$$1 - \binom{2n}{2} + \binom{2n}{4} - \binom{2n}{6} + \cdots + (-1)^n \binom{2n}{2n} = 2^n \cos\left(\frac{n\pi}{2}\right).$$

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$